Predicting NBA Playoff Success:

A Python-based round predictor

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Summary

This project predicts NBA playoff success from regular-season data using 29 seasons (1997-2025) and 862 team-seasons. We tested whether advanced statistics and machine learning could improve on our Fall 2025 Excel model. Win percentage alone explains 58% of variance in cross-validation. Adding engineered features and XGBoost raises cross-validated R^2 to 72%, a gain of 13.8 points. The remaining 28% reflects factors such as playoff injuries, matchup dynamics, and small-sample variance that regular-season data does not capture. On the 2025 holdout season the model reached $R^2 = 0.559$, predicting 53% of teams exactly and 87% within one round; it placed Oklahoma City, the eventual champion, in the Finals and picked Cleveland to win.

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Exhibit 1: Playoff Round Distribution (Histogram)

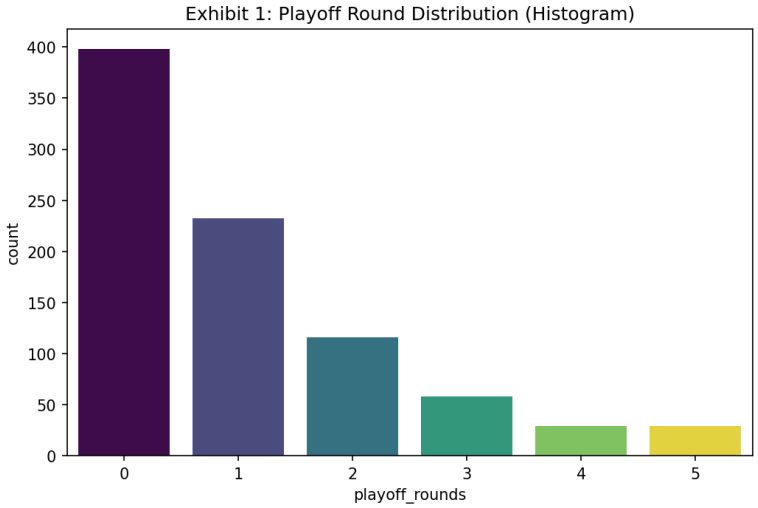


Exhibit 2: Correlation Matrix

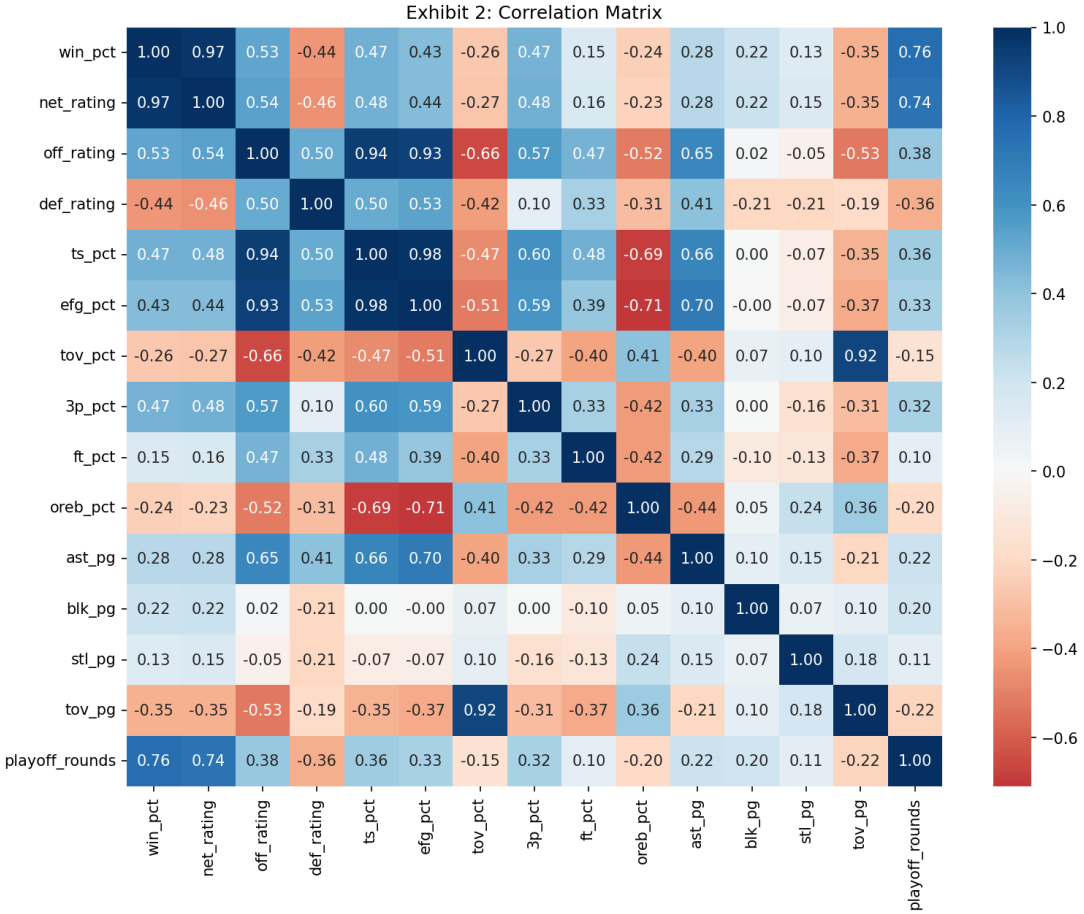


Exhibit 3: 3P% vs. Playoff Rounds (Scatterplot)

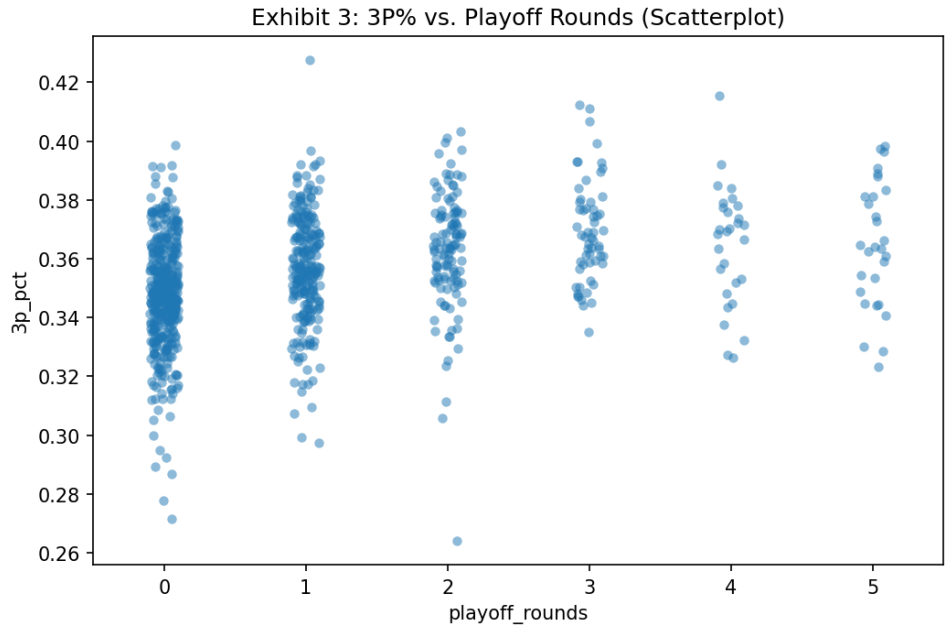


Exhibit 4: TS% vs. Playoff Rounds (Scatterplot)

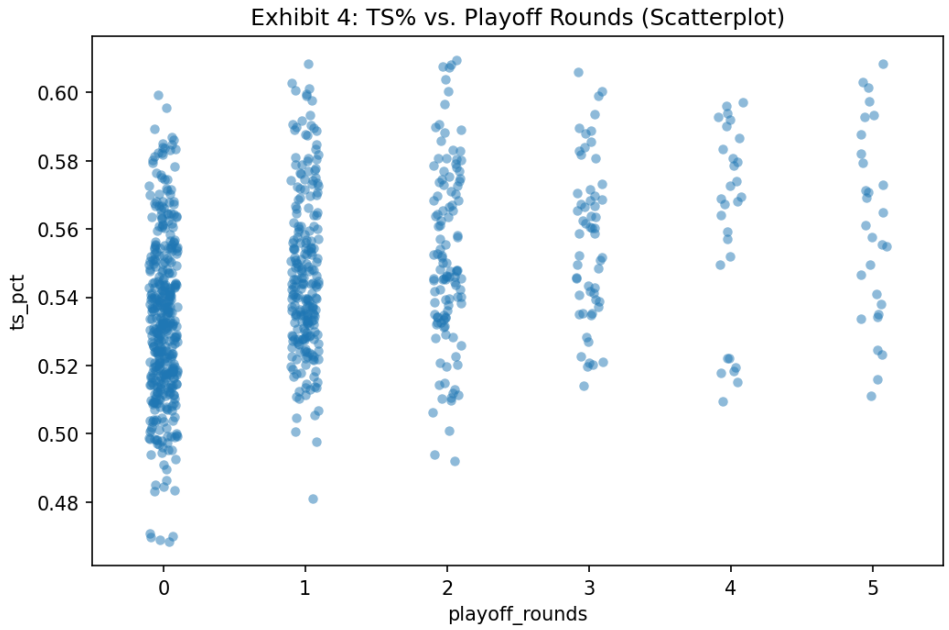


Exhibit 5: Net Rating vs. Playoff Rounds (Scatterplot)

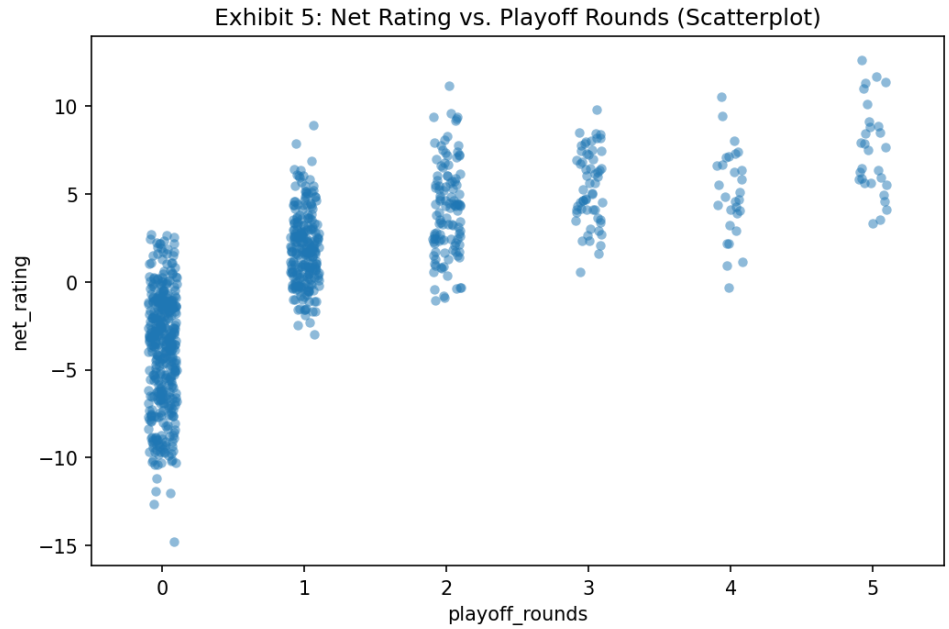


Exhibit 6: Regular Season Injuries vs. Playoff Injuries (Scatterplot)

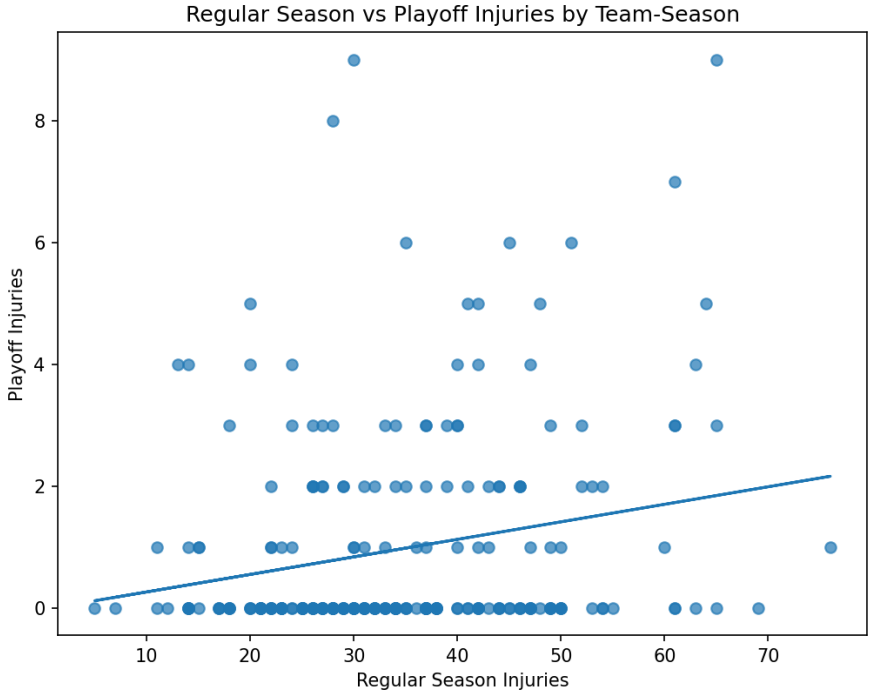
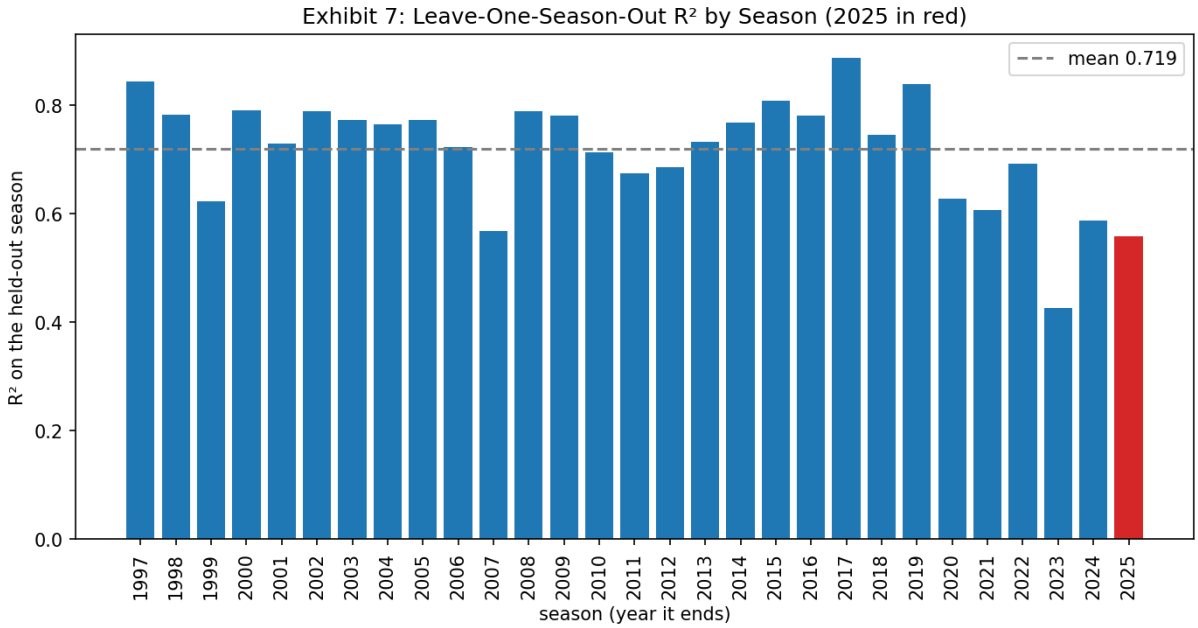


Exhibit 7: Leave-One-Season-Out R² by Season



References

[1] Moore, E. A. (2026). NBA Dataset: Box Scores and Stats (1947 - Today). Kaggle. https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores

[2] Lauton, L. (2023). NBA Player Injury Stats (1951-2023). Kaggle. https://www.kaggle.com/datasets/loganlauton/nba-injury-stats-1951-2023

[3] Hopkins, G. (2020). NBA Injuries 2010-2020. Kaggle. https://www.kaggle.com/datasets/ghopkins/nba-injuries-2010-2018

[4] Pastorello, G. (2025, April 24). Predicting the NBA champion with machine learning. *Towards Data Science*. https://towardsdatascience.com/predicting-nba-champion-machine-learning/

Introduction

The NBA markets itself as a league of parity, but underneath the upsets and storylines, there are patterns—beauty in the chaos. Certain teams consistently overperform in the playoffs, while others fall short of expectations time and time again. Analysts, front offices, and fans alike wonder: which team statistics matter most, and which ones only look good on paper?

In fall 2025, two of us built a rudimentary playoff predictor in Excel using eleven seasons of data, from 2014-15 through 2024-25, to test whether the modern NBA's obsession with three-point shooting was justified. The model achieved an R^2 of 0.52, confirming that Net Rating, a measure of point differential per 100 possessions, was the strongest predictor of playoff success. Three-point percentage mattered, but only until True Shooting absorbed its signal. The findings were simple; teams that consistently outscore opponents win in the playoffs.

However, that project left questions unanswered. Eleven seasons is a small sample for a league that has existed for 79 years. Excel, while accessible, lacks the statistical rigor and machine learning tools required for serious predictive modeling. Most importantly, an R^2 of 0.52 means nearly half of playoff outcomes remained unexplained. How close were we to the ceiling?

This project expands that foundation. We rebuilt the model in Python using 29 NBA seasons, from 1997 through 2025, covering 862 team-seasons across all 30 franchises. We replaced ordinary least squares with XGBoost, a gradient boosting algorithm designed to capture nonlinear relationships and feature interactions. We added contextual variables that the Excel model ignored: playoff seeding, late-season momentum, and regular-season injuries. These variables allowed us to account for who teams faced in the first round, how they performed in their final 20 games, and whether injuries during the regular season served as a proxy for depth or fragility.

Win percentage alone explained 58% of the variance in cross-validation. Adding 13 more team statistics raised cross-validated R^2 to 0.677, and adding playoff seeding and late-season form raised it to 0.719. In total, the engineered features added 13.8 points of R^2 over win percentage alone, while regular-season injury counts added almost nothing (0.001). The final model, trained on 1997-2024 and tested on the 2025 season, achieved a cross-validated R^2 of 0.719 and a holdout R^2 of 0.559.

We could not predict the remaining 28% of playoff variance from regular-season data. It likely reflects playoff injuries, which cannot be known in April; matchup-specific dynamics, such as defensive schemes that exploit particular offenses; small-sample randomness, since seven-game series are inherently noisy; and in-series coaching adjustments. These factors mark the limits of what regular-season performance can tell us about playoff outcomes.

This paper measures how much of playoff success regular-season data can predict. We test three hypotheses in sequence: first, that win percentage captures most of the predictable signal; second, that advanced statistics add marginal value; and third, that machine learning outperforms linear models. We validate the model on the 2025 season, placing Oklahoma City, the eventual champion, in the Finals but picking Cleveland to win, and missing hot playoff teams like Indiana. We explain why injuries, despite their intuitive importance, add almost no predictive power when measured only during the regular season. Regular-season durability does not necessarily predict playoff durability. Finally, we derive a composite Playoff Predictor Index that ranks teams within each season, achieving 53% exact accuracy and 87% within-one-round accuracy.

The goal is not only to produce the highest R^2 possible. It is to tell a coherent, honest story about the limits of prediction in a league where variance matters more than we want to admit. The playoffs, as they should be, remain partially unknowable.

Statement of Hypotheses

Hypothesis 1: Win Percentage is Effective

Among all regular-season statistics, win percentage is the strongest predictor of playoff success and will capture the majority of explainable variance.

Win percentage is the ultimate summary statistic. It encodes offense, defense, clutch performance, coaching, depth, and the intangible factors that separate good teams from great ones. A 60-win team executed consistently over 82 games. If any single metric should predict playoff success, it is the one that directly measures winning. We expect win percentage alone to explain the largest portion of playoff variance, setting the baseline.

Hypothesis 2: Advanced Statistics Add Marginal Value

Advanced efficiency metrics, including Net Rating, True Shooting, and pace-adjusted measures, will improve R^2 modestly but not dramatically.

Advanced statistics are often derivatives of win percentage. Net Rating measures point differential per possession. True Shooting measures scoring efficiency. These metrics decompose how teams win, not whether they win. A team with a +10 Net Rating almost certainly has a strong win percentage, so much of the explanatory power overlaps.

We expect 5 to 10 percentage points of improvement. Advanced statistics should refine the model by capturing nuance, such as the difference between dominant wins and ugly wins, but they probably will not transform it.

Hypothesis 3: Machine Learning Outperforms Linear Models

XGBoost will outperform linear regression by capturing nonlinear relationships and interactions.

Linear regression assumes predictors contribute independently. Basketball is interactive. Strong Net Rating may matter more if a team is a high seed and therefore has an easier first-round matchup. Late-season momentum might predict success only for elite teams. XGBoost can learn these patterns naturally.

However, we do not expect XGBoost to shatter the ceiling. The remaining 40% to 45% is not hiding in feature interactions. It is driven by factors absent from regular-season data: playoff injuries, opponent-specific matchups, and seven-game variance. No algorithm can predict what it has not seen.

Data Collection and Description

All data was sourced from Eoin Moore's Kaggle NBA dataset [1], covering NBA statistics from 1997-2025. For each team-season, we collected regular-season box scores from TeamStatistics.csv, game results from Games.csv, and player statistics from PlayerStatistics.csv. Injury data was merged from two Kaggle sources [2,3]: NBA Injury Stats (1951-2023) and NBA Injuries (2016-2025). Team names were standardized and mapped to unique identifiers.

Dependent Variable

Playoff rounds reached (Playoff_Rd) is an ordered discrete variable from 0 to 5:

0 = Missed playoffs

1 = First-round exit

2 = Conference semifinals exit

3 = Conference finals exit

4 = Finals loss

5 = Champion

Across 862 team-seasons, the distribution is heavily skewed: 398 zeros, representing non-playoff teams, and 232 ones, representing first-round exits, tapering down to 29 fours and 29 fives (Exhibit 1). Most teams do not experience deep playoff success, making rare outcomes statistically challenging to predict.

Independent Variables:

Team Performance Features

Win Percentage (win_pct)

Net Rating (net_rating): point differential per 100 possessions

Offensive and Defensive Rating (off_rating, def_rating)

True Shooting % (ts_pct): Points / [2 × (FGA + 0.44 × FTA)]

Effective FG % (efg_pct): (FGM + 0.5 × 3PM) / FGA

Turnover % (tov_pct)

3-Point % and Free Throw % (3p_pct, ft_pct)

Offensive Rebound % (oreb_pct)

Assists, blocks, steals, and turnovers per game

Contextual Features

1. Playoff Seed (playoff_seed): conference-based seeding from 1 to 8, or 0 for non-playoff teams. The top 8 teams per conference were ranked by win percentage. Higher seeds face easier first-round matchups.

2. Late-Season Form (late_form): Net Rating in the last 20 games minus season average. This measures whether teams peaked or declined heading into the playoffs.
3. Regular-Season Injuries (reg_season_injuries): injury count from October through March. This serves as a proxy for team depth and fragility.

All predictors were standardized to z-scores, with mean = 0 and standard deviation = 1, to allow coefficient comparison. Standardized variables are denoted with the ADJ prefix.

Statistics Summary

Win percentage ranges from 0.11 for tanking teams to 0.89 for elite championship contenders. Net Rating spans from approximately -15 for the league's worst teams to +13 for the 2025 Thunder. True Shooting generally hovers between 55% and 60%, while three-point percentage ranges from the low 30s to around 40%.

The correlation matrix (Exhibit 2) reveals several key patterns:

Win% and Net Rating: r = 0.97, showing that teams that outscore opponents almost always win games.
TS% and 3P%: r = 0.60, representing approximately 37% shared variance.
Net Rating with TS%: r = 0.48.
Net Rating with 3P%: r = 0.48.

These correlations include one severe case of multicollinearity. Win percentage and Net Rating are nearly interchangeable (r = 0.97), so a regression that contains both cannot separate their effects, and their individual coefficients should not be read on their own. Net Rating is also exactly Offensive Rating minus Defensive Rating. TS% and 3P% are closely related, so 3P% should lose power once TS% enters the model. The other correlations stay below the r > 0.8 threshold.

Scatterplots confirm linear relationships with substantial noise. Three-point percentage versus playoff success is positive but weak (Exhibit 3). TS% shows a slightly stronger pattern (Exhibit 4). Net Rating demonstrates the clearest relationship (Exhibit 5): teams one standard deviation above the mean reach later rounds, while negative Net Rating teams rarely advance. The injury variable showed near-zero correlation with playoff success despite its intuitive appeal. Teams

with high injury counts were no less likely to succeed, suggesting regular-season durability does not predict playoff health.

Methodology

We used multiple regression as the primary framework, running nested models to test hypotheses sequentially.

Model 1: Win Percentage Alone

$$\text{Playoff_Rd} = \beta_0 + \beta_1(\text{win_pct}) + \varepsilon$$

This model tests Hypothesis 1. If win percentage captures the majority of the signal, it should explain 45% to 50% of playoff variance.

Model 2: Full Team Statistics

$$\text{Playoff_Rd} = \beta_0 + \Sigma\beta_i(\text{team_stats}_i) + \varepsilon$$

This model adds all 14 team performance features. It tests Hypothesis 2 by measuring whether advanced metrics add predictive value beyond win percentage.

Model 3: Contextual Features

$$\text{Playoff_Rd} = \text{Model 2} + \beta_{15}(\text{playoff_seed}) + \beta_{16}(\text{late_form}) + \beta_{17}(\text{reg_season_injuries}) + \varepsilon$$

This model adds seeding, momentum, and injuries. It tests whether context improves prediction beyond season-long team statistics.

Validation

Cross-validation: We used 5-fold cross-validation. The dataset was split into five parts. Each fold trained on four parts and tested on the fifth. Average R^2 across folds provides a more honest estimate of model performance. The model was trained on 1997-2024 and tested on the 2025 season, which the model had never seen. This simulates real-world forecasting.

Hypothesis 3 compares XGBoost to linear regression. XGBoost builds an ensemble of decision trees, with each tree correcting errors made by the previous trees. Unlike linear regression, it can

capture nonlinearities, such as Net Rating having a stronger effect at the extremes, and interactions, such as high Net Rating mattering more for top seeds.

Hyperparameters were tuned to prevent overfitting:

> max_depth = 2, n_estimators = 500, learning_rate = 0.01, reg_alpha = 1.0, reg_lambda = 5.0

The same cross-validation procedure was applied to both models for direct comparison.

Recent seasons were weighted slightly higher using:

> $w = \exp(-0.01 \times \text{years_ago})$

This produces minimal decay, with 1997 weighted 0.76 relative to 2025. The goal was to reflect stable basketball fundamentals while still acknowledging changes in league style over time.

Results

Model 1

> $R^2 = 0.581$ (cross-validated)

Win percentage alone explains 58.1% of playoff variance. Its standardized coefficient is approximately 1.00 ($p < 0.001$). A one-standard-deviation increase in win percentage, roughly 12 to 13 more wins, predicts an increase of about 1.0 playoff round.

This baseline is strong for a single predictor. Roughly half of playoff variance is attributable to regular-season wins. Residuals are largest at the extremes, such as high-win teams that exit early due to injuries or lower-seeded teams that overperform. But on average, more wins mean deeper playoff runs.

Model 2

> $R^2 = 0.677$ (cross-validated); weighted OLS in-sample $R^2 = 0.588$, adjusted $R^2 = 0.582$

The gain over the baseline model was 9.6 percentage points. In the weighted OLS fit, win percentage remained the dominant predictor (1.00, $p < 0.001$). Net Rating's coefficient was small

and not significant (-0.09, p = 0.62), because win percentage and Net Rating are almost perfectly correlated (r = 0.97) and the regression cannot separate them. True Shooting had a positive but non-significant coefficient (0.16, p = 0.75). Three-point percentage had a small negative coefficient (-0.11, p = 0.013) once TS% and eFG% were in the model. This does not mean three-point shooting hurts. TS% and eFG% already capture most of that signal, so 3P% only picks up what is left over.

Steals, blocks, and assists showed small and inconsistent effects. Turnover percentage had a small positive coefficient that was not significant (0.10, p = 0.66).

The key takeaway is that advanced statistics add modest value. Moving from 1 to 14 predictors improved R^2 by 9.6 percentage points. This improvement does not scream meaningful, but marginal predictive power is better than a blind guess, and in the sports-analytics world, that matters.

Model 3

$R^2 = 0.719$ (cross-validated); weighted OLS in-sample $R^2 = 0.595$, adjusted $R^2 = 0.588$

Adding contextual features improved the model by 4.2 percentage points. Playoff seeding was significant in the weighted OLS fit (-0.09, p = 0.004). Seeds run from 1 (best) to 8, and non-playoff teams are coded 0, so the sign of this coefficient is not a clean measure of seed strength. Late-season form had a small negative coefficient that was not significant (-0.03, p = 0.37). Regular-season injuries had a small positive coefficient in-sample (0.06, p = 0.045), but adding them raised the cross-validated R^2 by only 0.001 (0.719 to 0.720), so they add almost no predictive value out of sample. They also had little correlation with playoff injuries, which we sourced from the rest of the Kaggle data (Exhibit 6). Injury count from October through March does not predict playoff success.

The overall fit was strong. A cross-validated R^2 of 0.719 means the model explains about 72% of playoff variance in seasons it was not trained on. Because every fold is tested on whole seasons the model never saw, this estimate is not inflated by fitting the test seasons. The remaining 28% is likely not explainable from regular-season data alone.

Linear Regression vs. XGBoost

Linear regression (17 features): $R^2 = 0.589$

XGBoost (17 features): $R^2 = 0.720$

XGBoost outperformed linear regression by 13.1 percentage points, confirming Hypothesis 3. Nonlinearities and interactions matter. XGBoost can learn patterns linear regression cannot.

Even so, about 28% of playoff variance remains unexplained. A cross-validated R^2 of 0.719 is the best result we reached with regular-season data, which supports the conclusion that the rest depends on information the data does not contain.

2025 Holdout Test

The model was trained on 1997-2024 and tested on 2025.

Holdout $R^2 = 0.559$

Exact prediction accuracy: 16/30 teams (53%)

Within-one-round accuracy: 26/30 teams (87%)

The model picked Cleveland as champion and Oklahoma City as the Finals loser; Oklahoma City won the title, and Cleveland lost in the conference semifinals. Its next two teams were Boston, which reached the conference semifinals, and Houston, which lost in the first round. It predicted the exact round for Denver, Milwaukee, Memphis, and Detroit, and correctly assigned zeros to 12 of 14 non-playoff teams.

One of the two biggest misses was the Indiana Pacers. The model predicted a first-round exit, but they reached the Finals. This is a classic hot playoff team problem. Indiana's regular-season statistics were good but not elite, with a Net Rating around +2 and the fourth seed in the East. The model could not know they would catch fire in the playoffs. The model also overestimated the LA Clippers, predicting a conference semifinals appearance when they exited in the first round. This was likely affected by playoff injuries.

The holdout R^2 of 0.559 is well below the cross-validated R^2 of 0.719 and below all five cross-validation folds (0.686 to 0.741), so the two should not be read as matching. A single season of 30 teams is a much noisier test than a fold of five or six seasons. To see how much one season can vary, we also held out each of the 29 seasons in turn and trained on the other 28 (Exhibit 7).

Single-season R^2 ranged from 0.425 (2023) to 0.887 (2017), with a mean of 0.719 and a median of 0.745. The 2025 holdout is the same test as the 2025 fold, and at 0.559 it is the second-lowest of the 29 seasons. Every season from 2020 through 2025 falls below the 29-season mean, so the model is less accurate on recent seasons than its average across the full window suggests.

Discussion

Win Percentage Dominates

Hypothesis 1 holds. Win percentage explains 58.1% of playoff variance, capturing approximately 81% of the total predictable signal (58.1 / 71.9). This confirms what coaches already know: the best predictor is winning games.

The practical implication is simple. When evaluating contenders, the first question should not be, "What is their three-point percentage?" It should be, "How many games do they win?" A 55-win team is usually a contender. A 45-win team is usually a fringe playoff team. A 35-win team is usually rebuilding. Exceptions exist, but they are rare and often driven by factors models cannot capture.

Advanced Statistics Add Modestly

Hypothesis 2 also holds. Adding the engineered features improved cross-validated R^2 from 58.1% to 71.9%, a gain of 13.8 percentage points. On its own, Net Rating is the second-best single predictor (cross-validated $R^2 = 0.545$), but it is so closely tied to win percentage ($r = 0.97$) that the regression cannot credit it with value beyond winning. Net Rating measures how much teams outscore opponents per possession, making it more granular than win percentage. There is a meaningful difference between teams that win close games and teams that consistently win by large margins. The latter is more sustainable in the playoffs, where variance averages out over seven-game series more than it does in single games. True Shooting's added value was not statistically significant. Three-point percentage contributed little once TS% entered the model. This does not mean three-point shooting is irrelevant. TS% is simply a better overall efficiency summary because it includes threes, twos, and free throws. A team that lives at the rim and free throw line can be as efficient as a perimeter-oriented team. TS% reflects that.

The implication for an analytics dashboard is clear: track Net Rating obsessively, use True Shooting as the primary scoring efficiency metric, and treat isolated three-point percentage as context rather than a standalone answer.

What Regular-Season Data Cannot Explain

Hypothesis 3 holds. XGBoost outperformed linear regression (0.720 vs. 0.589), confirming that nonlinearities and interactions exist. The 2025 holdout R^2 of 0.559 was lower than the cross-validated R^2 of 0.719, and the leave-one-season-out results show that recent seasons have been harder to predict than earlier ones.

The remaining 28% is driven by factors absent from regular-season data. Playoff injuries are the clearest example. Regular-season injury counts measure durability, but playoff success depends on playoff health exclusively. A star going down in May or June changes everything. The 2019 Warriors lost Kevin Durant and Klay Thompson in the Finals and went from favorites to losers. No regular-season stat can predict that specific event.

Matchup dynamics also matter. Some teams match poorly against specific opponents because of defensive schemes, pace, or style. Regular-season data averages performance across the league. The playoffs magnify one opponent over as many as seven games. A defense that stops pick-and-roll actions might dominate one opponent and struggle against another. Small-sample variance is another limitation. Seven-game series are noisy. One hot shooting night, one bad call, or one late-game mistake can swing a game and, eventually, a series. Over 82 games, variance has time to average out. Over seven games, it does not. In-series adjustments also matter. Coaches make mid-series strategic changes that are invisible in regular-season data. A team that struggled against zone defense in February might solve it by Game 4 of a playoff series.

These are not modeling failures. They are limits of what regular-season data can tell us. No algorithm can predict the genuinely unpredictable.

Comparison to Prior Work

Excel model (Fall 2025): in-sample $R^2 = 0.52$, 11 seasons

Python model: cross-validated $R^2 = 0.72$, 29 seasons

The key difference is not just the R^2 value. It is rigor. The Excel model used in-sample testing, which inflates performance by fitting noise. This model uses cross-validation and holdout testing, producing more honest generalization estimates. The Excel model tested three hypotheses and concluded that Net Rating matters most. Over a longer horizon, this project finds Net Rating nearly interchangeable with win percentage ($r = 0.97$), with win percentage slightly stronger on its own, and measures how much regular-season data can predict.

Implications for the Modern NBA

The data validates some conventional wisdom: three-point shooting matters. Teams that shoot well from deep score efficiently, win more, and usually have stronger Net Ratings. But the model also demonstrates the limitations of focusing on any single statistic. TS% is better than 3P% because it captures full scoring efficiency. Net Rating is better than TS% because it captures both offense and defense.

The biggest implication is humility. Analysts, front offices, and fans should resist overfitting narratives to regular-season statistics. A +10 Net Rating team is a title contender. A +3 Net Rating team is a playoff team. But whether the +10 team actually wins depends on injuries, matchups, and variance that no one can know in April.

Stephen Curry and the Warriors changed the geometry of the game. Three-point shooting is more valuable now than it was in the 1990s. But the fundamentals remain unchanged. What matters most now is the same as always: consistently outscoring opponents. Net Rating is not revolutionary. It is an evergreen truth.

Limitations and Future Work

Data Constraints

We lack playoff injury data, which is likely the single biggest driver of unexplained variance. If we had games missed during the playoffs for each team's top three players, R^2 would almost certainly improve. Injuries are the elephant in the room of playoff prediction, but they are also unknowable in advance.

We also lack player tracking data, such as speed, shot location, spacing, and defensive matchup metrics. These would capture nuances beyond traditional box scores. A team generating open threes is likely more sustainable than a team hitting contested threes, but traditional statistics cannot distinguish between the two.

We assume current East-West alignment applies to all 29 seasons, which is historically imperfect. Some teams switched conferences, and the playoff structure evolved, especially after the play-in tournament was added in 2021. This simplification was necessary for modeling, but it remains a limitation.

What Else Would Push R^2 Higher

1. Vegas betting lines: betting markets aggregate scouting, insider knowledge, and advanced models. Including preseason championship odds would likely improve R^2 substantially, but it would make the model less interpretable and less useful diagnostically.
2. Elo ratings: dynamic game-by-game ratings updated after every win and loss. These would capture momentum and recency better than season averages.
3. Player tracking: shot quality, defensive matchups, and spacing metrics.

Conclusion

We set out to improve our Fall 2025 Excel model by expanding the dataset, adding features, and using machine learning. We succeeded in building a more rigorous and better-validated model. The most important finding is how much of playoff success regular-season data can explain, and how much it cannot.

With regular-season performance, we were able to explain approximately 72% of playoff success in cross-validation. Win percentage alone captures 58%. The 13 additional team statistics add about 10 percentage points, and contextual features, such as seeding and momentum, add about 4 more, for a total gain of 13.8 points over win percentage. Machine learning captures nonlinearities and interactions that linear models miss, adding about 13 points over linear regression. Regular-season injury counts add almost nothing.

The other 28% comes from playoff injuries, matchup dynamics, seven-game variance, and in-series adjustments.

Comparable public work reached a similar but not identical result. Pastorello's (2025) LightGBM "Champion Share" model, trained on 40 seasons of regular-season inputs, reached a test R^2 of 0.537 [4]. Our cross-validated R^2 of 0.719 is higher, and our single-season holdout R^2 of 0.559 is close to his figure, but the comparison is loose: the models predict different targets, cover different seasons, and are validated in different ways.

The model is not perfect, but it is honest.